

SignX

Text-to-3D Motion Generation for American Sign Language (ASL)

1. Executive Summary

Communication is a fundamental human right, yet the digital world remains largely inaccessible to over **430 million deaf and hard-of-hearing individuals** worldwide. SignX is an end-to-end Machine Learning system designed to bridge this digital divide. By translating natural English text into continuous, culturally authentic 3D American Sign Language (ASL) animations, SignX provides a scalable, automated alternative to human translation. This project lays the architectural foundation for accessible digital avatars, real-time media translation, and metaverse integration.

2. The Problem Statement

The current standard for digital accessibility is written text or closed captions. However, this approach has significant structural flaws:

- **Text is a Second Language:** Sign languages are not merely spoken languages expressed with hands—they possess entirely distinct grammatical structures, syntax, and spatial rules. For many native signers, reading written text is equivalent to reading a second language.
- **The Scalability Bottleneck:** Integrating sign language into digital platforms currently requires recording human interpreters in a studio—a process that is slow, expensive, and impossible to scale for real-time dynamic content.
- **Loss of Nuance:** Plain text lacks the emotional tone, urgency, and spatial context that is effortlessly conveyed through a signer's facial expressions and body kinematics.

3. The Solution: The SignX Engine

SignX leverages advanced deep learning to act as an automated text-to-3D-motion translator. Instead of relying on pre-recorded 2D videos, the system generates raw mathematical movement. By training on the **How2Sign Holistic** dataset, the model learns the spatial relationships of human joints (extracted via Google Mediapipe). It uses a sequence-to-sequence Transformer architecture to map semantic text embeddings directly to continuous 3D spatial coordinates—animating the body, hands, and face frame-by-frame.

4. Full-Stack System Architecture

SignX operates as a cohesive, three-tier architecture spanning the frontend, backend, and AI core.

Tier	Key Components
Frontend (UI)	Text input module, Three.js / React Three Fiber WebGL viewer, export controls (replay, speed, MP4 download)
Backend (API)	FastAPI server for high concurrency, PyTorch ML pipeline integration, coordinate smoothing and JSON formatting
ML Pipeline (AI Core)	DistilBERT semantic encoder, length estimation network, Spatial-Temporal Transformer Decoder, MSE loss optimization

A. The User Interface (Frontend)

A minimalist, highly responsive web application built to provide a seamless user experience.

- **Input Module:** A clean text area accepting natural English sentences.
- **3D Viewer:** An integrated WebGL canvas (using Three.js / React Three Fiber) that renders generated Mediapipe spatial coordinates onto a skeletal rig in real-time.
- **Export Controls:** Options to replay, adjust playback speed, and download the generated animation as an MP4 video file.

B. The Inference API (Backend)

A high-performance API designed to handle the computational load of 3D motion generation.

- **Framework:** Built with FastAPI for high concurrency, receiving text payloads and routing them to the PyTorch pipeline.
- **Coordinate Processor:** Smooths raw mathematical coordinates and formats them into a JSON payload consumable by the frontend 3D engine.

C. The Machine Learning Pipeline (AI Core)

The PyTorch-based neural engine that drives the translation operates in four sequential phases:

- **Phase 1 — Semantic Encoding:** DistilBERT extracts deep contextual meaning from input text, converting it into a dense mathematical embedding.
- **Phase 2 — Length Estimation:** A neural network predicts the temporal duration (frame count) required for the specific sign language sequence.
- **Phase 3 — Continuous Motion Generation:** A Spatial-Temporal Transformer Decoder predicts continuous (X, Y, Z) coordinates for body, hands, and face at every frame.
- **Phase 4 — Optimization:** MSE loss training forces generated skeletal movements to match ground-truth How2Sign motion capture data.

5. Project Roadmap & Future Scope

- **Current Phase:** Model architecture development and training on the How2Sign Holistic 3D dataset.
- **Next Steps:** Developing the React frontend and mapping Mediapipe landmark outputs to a fully rigged 3D human avatar.
- **Future Vision:** Implementing real-time speech-to-sign capabilities (processing live audio into 3D motion) and expanding dataset support to global dialects including British Sign Language (BSL) and Sri Lankan Sign Language.